

Solution Requirements

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Functional Requirements (FR) define what the system should do. Non-Functional Requirements (NFR) define how well it should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register and log in with email and password. Passwords are hashed and a JWT access token is issued and verified by middleware on every protected route.	High
2	Authorization levels	Two roles: User (manages only their own profile, logs, goals and reports) and Admin (views and manages all users and data).	High
3	External interfaces	Exposes a REST API (JSON) to clients such as Postman or any web/mobile app; stores data in MongoDB through Mongoose; calls an external AI service over HTTP for recommendations.	High
4	Transactions processing	CRUD for workouts, meals, water intake, goals and profile. Each request is validated, saved to MongoDB and returned with a status code and a standard JSON response.	High
5	Reporting	Daily, weekly and monthly summaries: calories, workout duration, streaks, goal progress and AI recommendation history, returned as JSON.	Medium
6	Business rules, etc.	A user can only access their own records. Required fields and value ranges are validated (e.g. positive weight and duration). AI plans use the user's profile, goal and recent logs.	High
7	Compliance to laws or regulations	Personal and health-related data is stored only for the owner and is never shared; basic data-privacy practices (consent, minimal data, deletion on request) are followed.	Medium
8	Other	Centralised error handling with meaningful messages; automated tests for key endpoints; .env-based configuration.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Standard CRUD endpoints respond quickly; AI recommendation calls are handled asynchronously.	CRUD responses < 500 ms; AI responses < 10 s

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
2	Scalability	Stateless Express server (JWT) and indexed MongoDB collections so the API can scale horizontally.	Supports 500+ concurrent users in testing
3	Security & Data Privacy	Hashed passwords (bcrypt), JWT authentication, input validation, secrets kept in .env, HTTPS when hosted.	No plain-text passwords; protected routes return 401 / 403 without a valid token
4	Reliability & Availability	Graceful error handling and database reconnect logic so one bad request does not crash the server.	99% uptime when hosted; no unhandled exceptions in tests
5	Usability & Accessibility	Consistent endpoint naming, clear JSON responses and error messages, documented in the README and a Postman collection.	All endpoints documented and callable from Postman
6	Other	Maintainability and portability through modular folders (routes, controllers, models, middleware, config) and a Node.js / npm setup that runs on any OS.	Setup completed with npm install and npm start